

Pavani Jain

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Education

Northeastern University

Master of Science in Artificial Intelligence | GPA: 3.54/4.00

Expected May 2027

Boston, MA

Guru Gobind Singh Indraprastha University

Bachelor of Computer Applications (BCA) | GPA: 3.64/4.00

Aug 2022 – Jun 2025

Delhi, India

Skills

Languages: Python, R, Java, C, C++, JavaScript, Bash, SQL, HTML, CSS

ML/AI: Deep Learning, NLP, Generative AI, LLMs, RAG, Agentic AI, AI Orchestration, MLOps, Anomaly Detection, Prompt Engineering, A/B Testing

Libraries: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, LangChain, LangGraph, HuggingFace, Transformers, OpenAI API, Anthropic API, SpaCy, PySpark

Data & Tools: PostgreSQL, pgvector, FAISS, MongoDB, BigQuery, Snowflake, ETL, Tableau, Power BI, FastAPI, AWS (SageMaker, S3, EC2, Bedrock), Docker, Kubernetes, Git, MLflow, CI/CD

Experience

Graduate Teaching Assistant – Information Retrieval

Sep 2026 – Present

Northeastern University

Boston, MA

- Support a 100+ student cohort in Information Retrieval, holding office hours, grading, and contributing to course design and assignments; explain ranking, embeddings, retrieval, and LLM concepts to students at varied levels.
- Review student code and communicate complex technical ideas clearly to technical and non-technical audiences.

AI Engineer Intern

May 2026 – Aug 2026

Thingr

Austin, TX

- Architected the AI orchestration layer (Python, LangChain, LangGraph, OpenAI/Anthropic APIs) powering 5+ multi-agent, multi-turn workflows with stateful agent graphs, 8+ tool-calling nodes, and structured output parsing.
- Built an LLM-as-judge evaluation pipeline scoring 6 competency dimensions over 50+ behavioral signals per session, with prompt-engineered guardrails, hallucination checks, and chain-of-thought traces for auditable scorecards.
- Developed a production RAG pipeline (FAISS, pgvector, hybrid retrieval) over 100+ scenarios at ~90% top-5 accuracy, deployed via FastAPI microservices and Docker on AWS (EC2, Bedrock) – cutting session spin-up from ~2 min to ~25 sec and P95 latency to ~1.8 sec.

Machine Learning Researcher – Financial Risk & Fraud Detection

Oct 2025 – Dec 2025

Northeastern University

Boston, MA

- Co-authored a research paper on autoencoder-based fraud detection on imbalanced data (<0.2% fraud), reviewing 25+ works and comparing 4 architectures (Logistic Regression, Random Forest, XGBoost, Autoencoder) with SMOTE, undersampling, and feature engineering.
- Ran end-to-end experimentation (preprocessing, threshold tuning, ablation studies) reaching 94% ROC-AUC, and documented metrics (ROC-AUC, precision-recall, F1) with reproducible results per ML research standards.

Data Analyst Intern

Jun 2024 – Sep 2024

Verve Bridge

Delhi, India

- Identified a ~6% payment success-rate gap by writing 50+ SQL queries (CTEs, window functions, joins) on 50K+ transactions, slicing KPIs by payment method, bank, and merchant segment – surfacing 3 underperforming routes that triggered an A/B test on retry logic.
- Built an end-to-end ETL pipeline in Python (Pandas, NumPy, SQLAlchemy) from PostgreSQL – cutting manual prep from ~3 hrs to ~20 mins with data-quality checks; delivered 3 Tableau dashboards adopted in weekly business reviews.

Projects

FairSearch – Retrieval Fairness Evaluation (RAG & LLMs)

May 2026 – Aug 2026

Python, LangChain

- Audited institutional-prestige bias in dense retrieval and LLM generation over a ~50K-paper arXiv corpus (team of 4); owned corpus enrichment and the fairness-evaluation harness (NDCG@10 0.56, MRR 0.70).
- Quantified precision/fairness trade-offs across ranking strategies; public on GitHub.

Nanochat – GPT Pretraining & LLM Systems Optimization

Mar 2026 – Apr 2026

Python, PyTorch, Transformers

- Built a GPT-style language model from scratch: BPE tokenizer, multi-head attention, training loop, and chat interface; trained on FineWeb-Edu with 8×H100 GPUs, reaching GPT-2-level validation loss in ~3 hrs at ~\$72 compute cost.
- Studied compute-quality tradeoffs by varying transformer depth, learning-rate schedules, and batch sizing.

Activities

1st Runner-Up, NU Hacks (48-hour hackathon)

March 2026

Northeastern University

Boston, MA

- Built a RAG clinical Q&A prototype with LangChain, indexing 1,000+ medical documents for cardiology/oncology queries; trained an LSTM on 500+ simulated patient timelines to flag early heart and cancer risk indicators.
- Optimized retrieval via ANN search and prompt tuning, achieving sub-2-second average query response time; placed 1st runner-up (team of 4).